Umm Al-Qura University

Data Management and Analysis Project

https://colab.research.google.com/drive/1EPJUB3DRK_ZgLAwpNXZJQUkjQrO-7bOT?usp=sharing

Members

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Introduction :

In this project, we aim to analyze video game sales data and identify the factors that influence global sales. The study is based on a dataset containing details about games, including platform, genre, publisher, and sales distribution across different regions. However, the dataset contains missing values and duplicates that need to be cleaned.

Dataset Description:

- Fields:
  - Game Title
  - Platform
  - Year
  - Genre
  - Publisher
  - Sales in North America
  - Sales in Europe
  - Sales in Japan
  - Sales in the Rest of the World
  - Global Sales
  - Ratings
  - Number of Rows: 1906
  - Number of Columns: 11
  - Source: Kaggle

Data Exploration :

- Data Cleaning: Missing values in categorical features were filled with 'Unknown', and missing years were filled with the median year. Duplicate entries were removed.
- Data Transformation: Categorical variables were converted to numerical values using label encoding. A new feature, "Age", was created by subtracting the release year from the current year (2025).
- Univariate Analysis: Histograms and box plots were used to visualize the distribution of numerical features. Log scaling was applied to reduce extreme differences.
- Bivariate/Multivariate Analysis: Pair plots were used to visualize relationships between features and the target variable.

Modeling process:

- Models Used

Linear Regression: The primary model used for predicting global sales.

- Features Selected

All features except the target variable 'Global' were used as input features.

- Hyperparameters Chosen

Default hyperparameters for the Linear Regression model were used.

Evaluation:

- Training Accuracy:

The model achieved a training accuracy of approximately 0.93.

- Testing Accuracy:

The model achieved a testing accuracy of approximately 0.92.

- Overall Accuracy:

The overall accuracy of the model on the test set was approximately 0.92

Data Cleaning:

```
[ ] #Handle missing values using (median)
    Videogame.fillna({
        'Publisher': 'Unknown',
        'Genre': 'Unknown',
        'Platform': 'Unknown',
        'Year': Videogame['Year'].median()  # Fill missing years with the median
    }, inplace=True)
```

```
[ ]  #Remove duplicate entries
    cleaned_data = Videogame.drop(columns=['index', 'Rank']).drop_duplicates()

     # Save the Cleaned Dataset
    cleaned_file_path = 'Cleaned_Video_Games_Sales.csv'
    cleaned_data.to_csv(cleaned_file_path, index=False)
```

```
[ ] #Convert categorical variables into numerical values (using label encoding).
    categorical_features = ['Game Title','Publisher', 'Genre', 'Platform']
    for feature in categorical_features:
        label_encoder = LabelEncoder()
        cleaned_data[feature] = label_encoder.fit_transform(cleaned_data[feature])
    cleaned_data.head()
```

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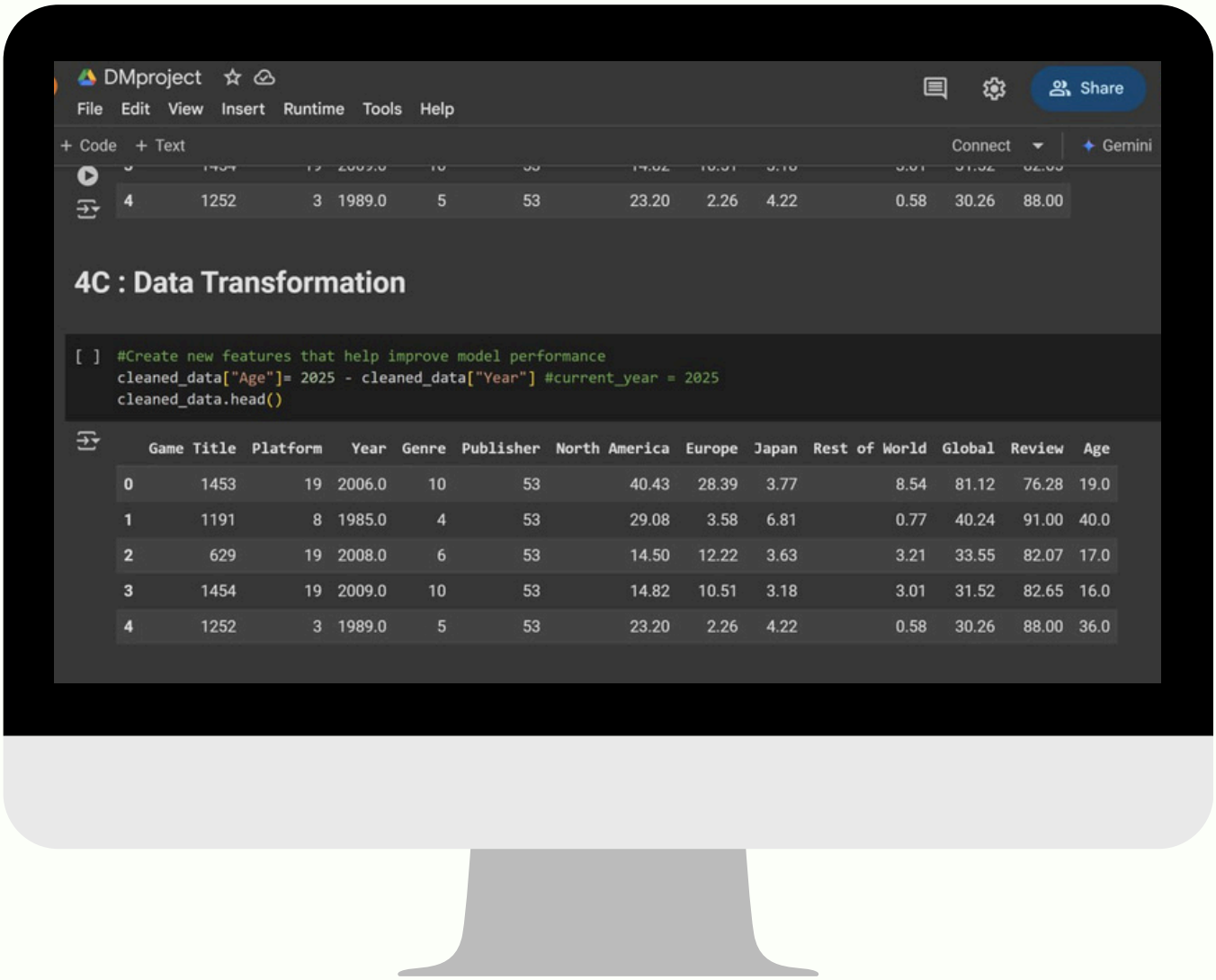
Data Transformation :



	Game Title	Platform	Year	Genre	Publisher	North America	Europe	Japan	Rest of World	Global	Review	Age
0	1453	19	2006.0	10	53	40.43	28.39	3.77	8.54	81.12	76.28	19.0
1	1191	8	1985.0	4	53	29.08	3.58	6.81	0.77	40.24	91.00	40.0
2	629	19	2008.0	6	53	14.50	12.22	3.63	3.21	33.55	82.07	17.0
3	1454	19	2009.0	10	53	14.82	10.51	3.18	3.01	31.52	82.65	16.0
4	1252	3	1989.0	5	53	23.20	2.26	4.22	0.58	30.26	88.00	36.0

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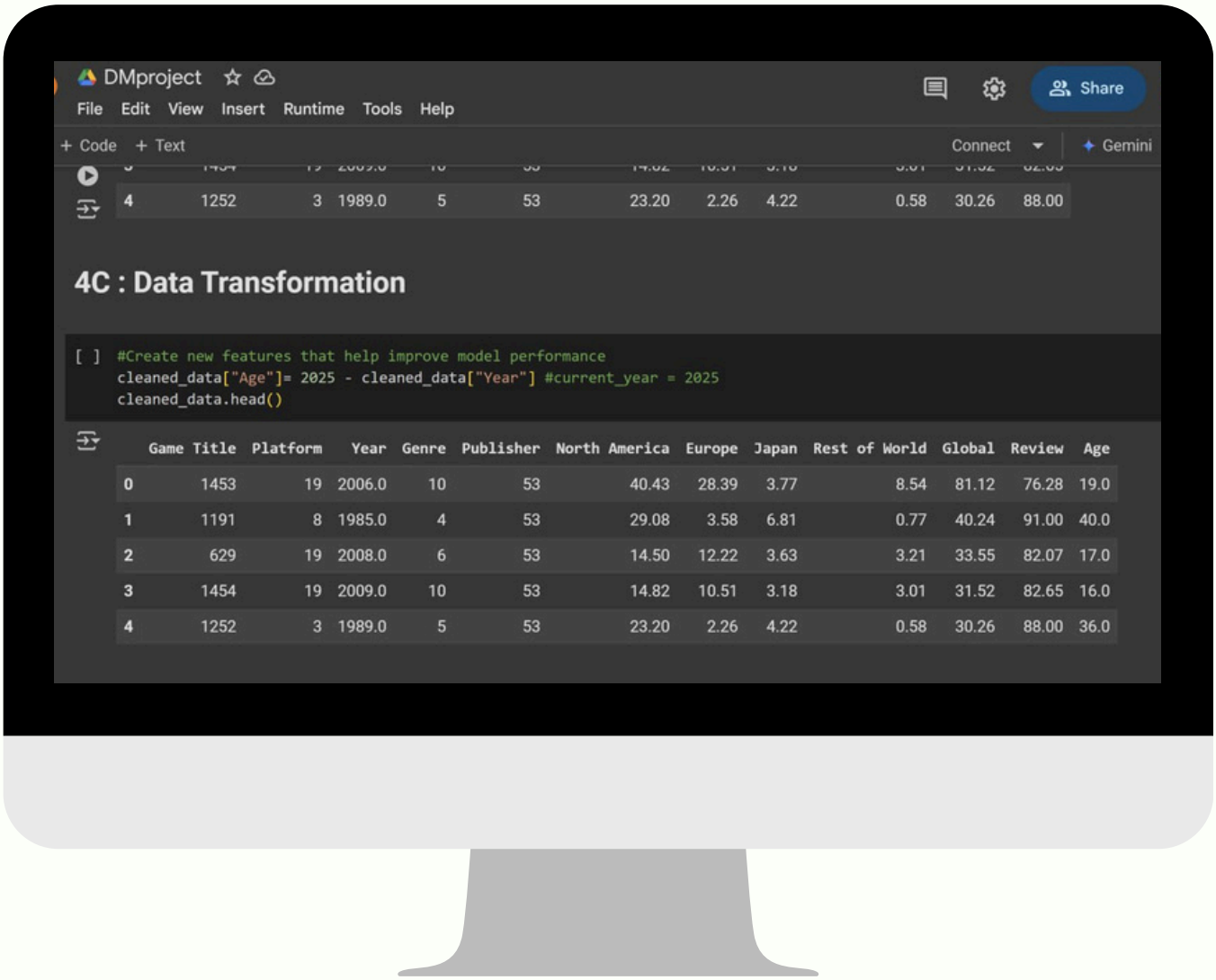

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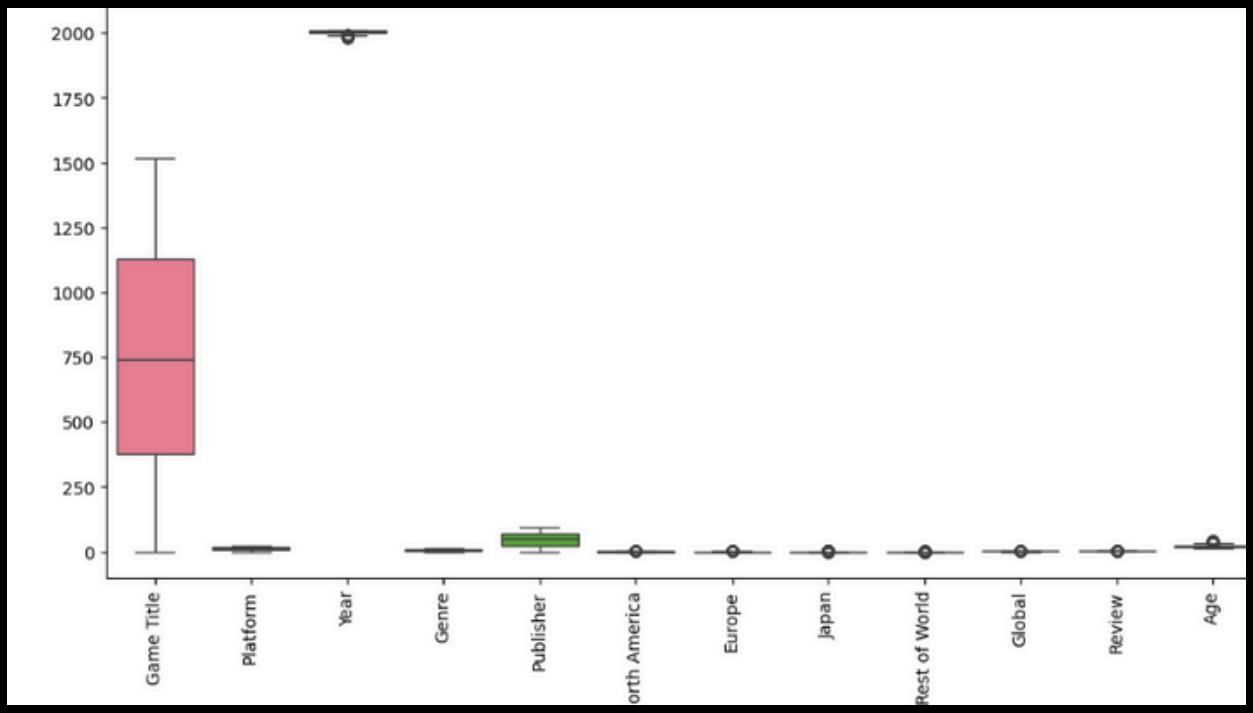

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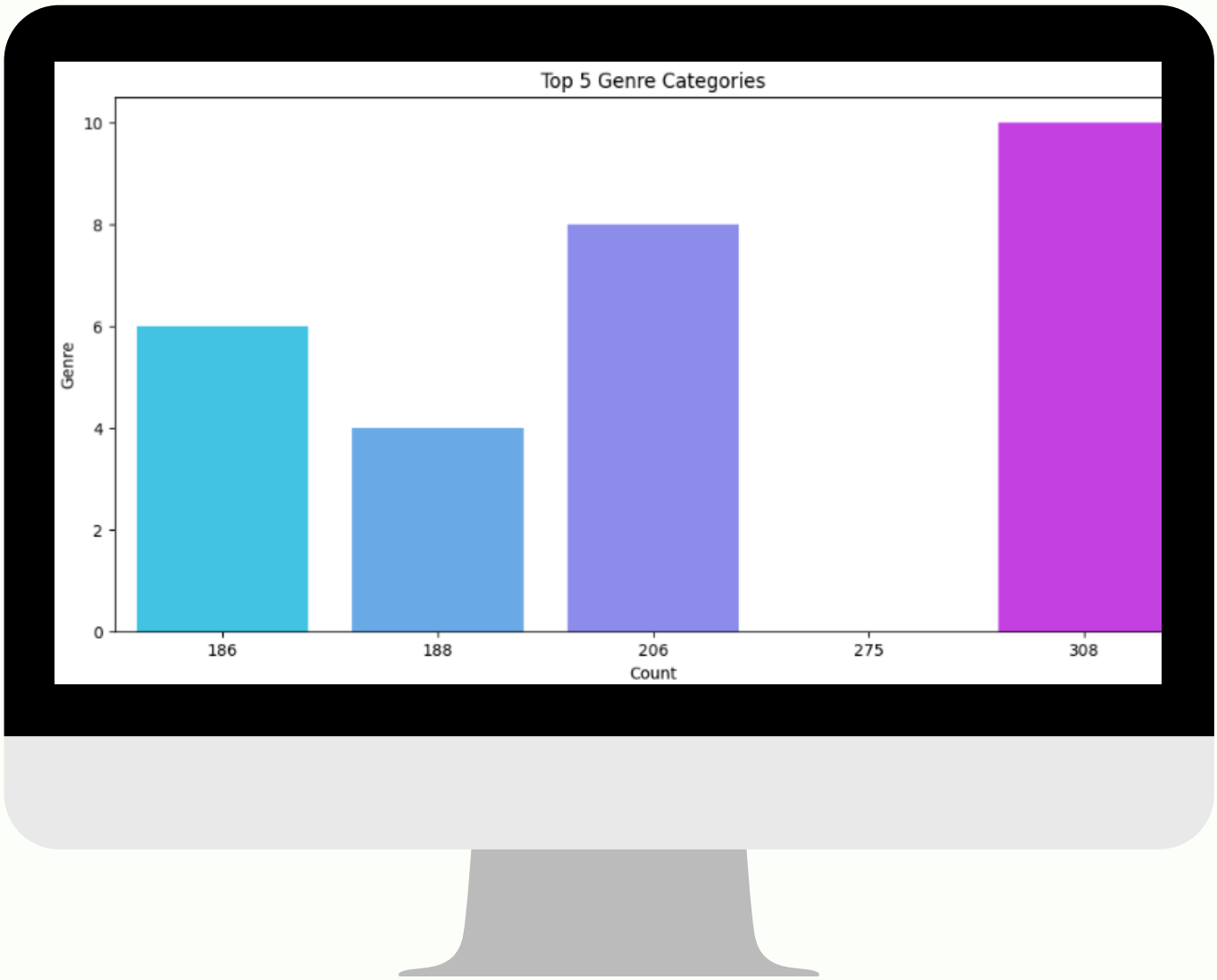
Visualizing the Distribution of the Target Variable and Features Histograms, box plots, and bar charts



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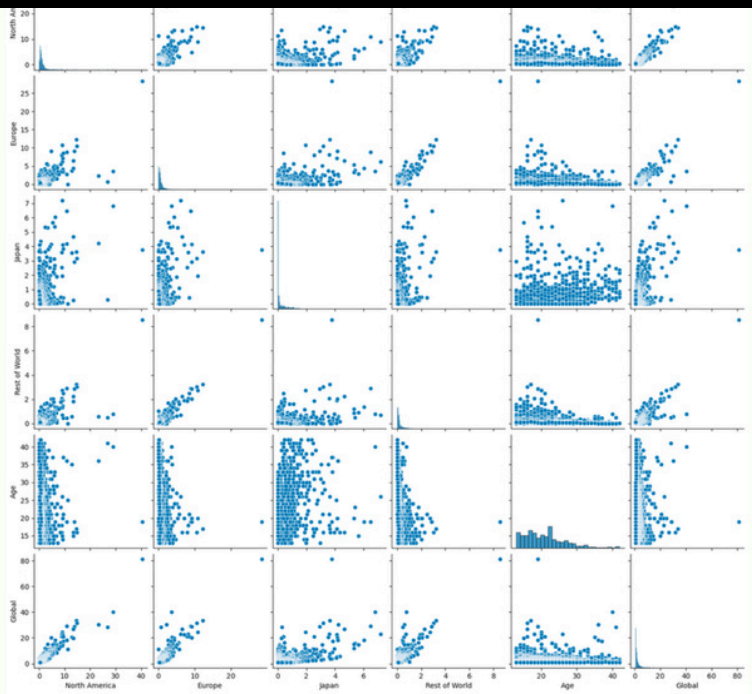
Multivariate Analysis :

Visualize relationships between features and the target variable (using pair plots)



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Conclusions:

- The Linear Regression model performed well, with high accuracy on both training and testing sets, indicating effective learning without significant overfitting or underfitting.
- The data cleaning and transformation steps, such as handling missing values and encoding categorical variables, were crucial in preparing the dataset for modeling.
- The creation of the "Age" feature helped improve the model's performance by providing additional context about the game's release year.
- Further improvements could be explored by trying different models, tuning hyperparameters, and incorporating additional features.
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References:

- Python libraries for data analysis (Pandas, NumPy, Scikit-learn)
- Global Video Game Performance:
- https://search.app/tmkx22fTPkCWNznR8
- https://pandas.pydata.org/docs/user_guide/missing_data.html